



Highest common factor

TASK A: Common factors using prime factorisation

Identify the incorrect common factor of the products given in the first column. There is one in each row.

$2^4 \times 3^2 \times 11$ and $2^2 \times 3^5 \times 11^2$	$2 \times 3 \times 11$	$2^2 \times 3^2$	$2^4 \times 3^5 \times 11^2$	$2^2 \times 11$
$3 \times 7^4 \times 13^3$ and $3^3 \times 7 \times 13^2$	$3 \times 7 \times 13^2$	$3^3 \times 7 \times 13^2$	$3 \times 7 \times 13$	7×13^2
$2^4 \times 5^2 \times 19^2$ and $3^4 \times 5^2 \times 19^5$	$5^2 \times 19^2$	5×19^2	$5^2 \times 19$	$2^4 \times 3^4 \times 5^2 \times 19^2$
$2^8 \times 3^5 \times 7^4 \times 11^4$ and $3^4 \times 7^2 \times 11^7$	$3^4 \times 7^2 \times 11^7$	$3^4 \times 7^2 \times 11^4$	$3^3 \times 7 \times 11^3$	$7^2 \times 11^2$
$2^4 \times 3^2 \times 5^3 \times 7^2$ and $2 \times 3^4 \times 5^4 \times 7^3 \times 11^2$ and $2^3 \times 5^4 \times 7^3 \times 11^3$	2×5^2	2×7^2	$2^4 \times 5^2 \times 7$	$5^2 \times 7^2$



TASK B: Highest common factor using prime factorisation

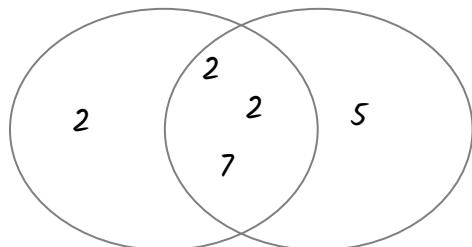
1) Find the **HCF** of the following pairs of numbers:

a) $56 = 2 \times 2 \times 2 \times 7$

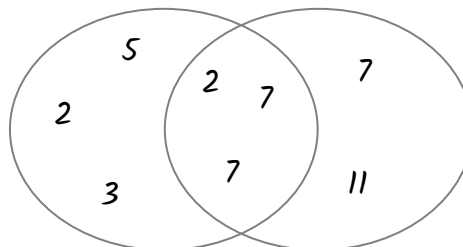
b) $2940 = 2 \times 2 \times 3 \times 5 \times 7 \times 7$

$140 = 2 \times 2 \times 5 \times 7$

$7546 = 2 \times 7 \times 7 \times 7 \times 11$

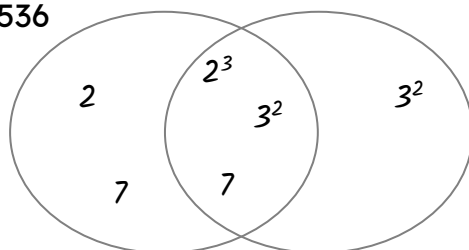


The **HCF** of 56 and 140 is 28



The **HCF** of 2940 and 7546 is 98

c) $7056 = 2^4 \times 3^2 \times 7^2$
 $4536 = 2^3 \times 3^4 \times 7$

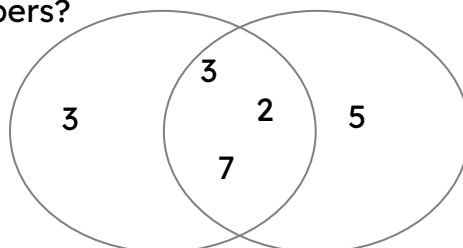


The **HCF** of 7056 and 4536 is 504

2) The **HCF** of the two numbers represented in the Venn diagram is 210.

a) What is the missing prime factor?

b) What are the two numbers?



a) $2 \times 3 \times 7 \times \boxed{5} = 210$

b) Left number is $2 \times 3 \times 3 \times 5 \times 7 = 630$, right number is $2 \times 3 \times 5 \times 5 \times 7 = 1050$

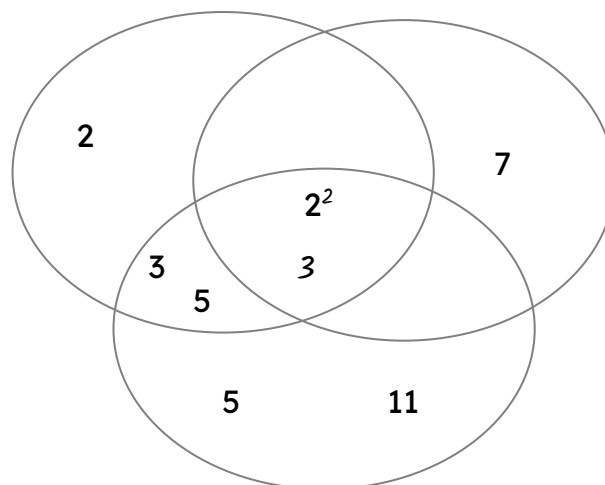
TASK C: Highest common factor of more than two numbers

Fill in the blanks

$360 = 2^3 \times 3^2 \times 5$

$84 = 2^2 \times 3 \times 7$

$9900 = 2^2 \times 3^2 \times 5^2 \times 11$



The **HCF** of 360, 84 and 9900 is 12

